

NEXUS-1 — Model Analysis Parameter Reference

A parameter-by-parameter explanation of every figure on the Model Analysis screen, organised in the same order as the screen itself: Solver validation · Model parameters · Operating margins · Sensitivity & uncertainty · Xenon forecast.

Honest scope

Every figure on the Model Analysis screen is computed live from the same point-kinetics solver the console runs (`window.KIN.model`) — nothing is hand-entered. The screen verifies the solver's numerical fidelity to the point-kinetics equations; it does not claim agreement with any specific real reactor. The constants are representative textbook-order values for the selected unit class (PWR-900, PWR-1300 or NUWARD-170), so values shown in the console depend on the unit selected. This document therefore explains what each parameter is and why it is there, rather than repeating unit-specific numbers.

1 · Solver validation — computed vs reference

Each row integrates the live solver against a closed-form reference and shows the genuine residual. The inhour and prompt-jump checks run with temperature and xenon feedback disabled, because that is the regime those closed forms describe.

Check	What it verifies, and why it matters
Steady state ($\rho = 0$)	With reactivity at zero, power must hold at 100% — the figure shown is the maximum drift over 60 s of integration. Purpose: proves the implicit numerical scheme is stable at equilibrium and does not self-excite. A drift below 0.05% passes.
Delayed fraction β (Σ of 6 groups)	The total delayed-neutron fraction, summed over the six precursor groups. Purpose: it is the single most important safety constant in kinetics — it defines \$1.00 of reactivity (β expressed in pcm) and the boundary of prompt criticality.
Asymptotic period, +10 pcm and +100 pcm	The solver is run with a small constant reactivity insertion and the resulting stable (asymptotic) period is measured by log-linear fit, then compared against the period predicted by the inhour equation for the same constants. Purpose: the inhour equation is the canonical analytic solution of point kinetics; agreement to a fraction of a percent demonstrates the integrator solves the equations it claims to solve. Two insertions are checked to cover both a slow and a faster transient.
Prompt jump, +200 pcm	Immediately after a reactivity step, power jumps almost instantly to approximately $\beta/(\beta-\rho)$ before the delayed groups take over. The solver's value 50 ms after the step is compared with that approximation. Purpose: verifies the fast (prompt) timescale of the solver, complementary to the slow asymptotic-period check.
I-135 / Xe-135 half-lives	The model's decay constants converted to half-lives, shown against the textbook values (≈ 6.57 h and ≈ 9.14 h). Purpose: transparency — these rows are labelled REPR. because any difference reflects the representative constants chosen, not a solver fault.
Post-shutdown xenon peak	A forward integration of the iodine/xenon balance from 100% equilibrium after a shutdown, reporting when the xenon peak occurs and how high it is relative to equilibrium. Purpose: the classic 7–11 h post-trip xenon build-up is one of the best-known qualitative behaviours of a thermal reactor; reproducing it shows the poison model has the right structure.
Equilibrium decay-heat fraction	The sum of the 12-group afterheat model at shutdown, compared with the canonical ≈ 6.5 –7% of rated power. Purpose: anchors the decay-heat model at $t = 0$.
Decay heat at $t+60$ s / 1 h / 24 h	The 12-group model evaluated analytically (shutdown from infinite operation) against the Way-Wigner correlation $P/P_0 = 0.0622 \cdot t^{-0.2}$. Purpose: shows the group fit tracks the standard engineering correlation across four decades of time; the Δ shown is the genuine residual of the fit.

Check	What it verifies, and why it matters
Thermal equilibrium (100%)	The two-node thermal model's analytic equilibrium temperatures (fuel and coolant) compared against the declared reference operating point. Purpose: a self-consistency check — the thermal gains, heat capacities and reference temperatures must describe the same operating point, or the feedback terms would be miscalibrated.

The small non-zero errors shown on screen are the real numerical-integration error of the live scheme, not cosmetic figures.

2 · Model parameters — single source of truth

The literal constants inside the running solver, deliberately split into two groups: physical nuclear data (evaluated quantities a licensed code would also use) and representative modeling choices (lumped or hand-tuned values that stand in for plant-specific engineering data). The split itself is the point: it tells a reviewer exactly which numbers are physics and which are modeling.

2.1 · Nuclear data (physical constants)

Parameter	Value / form	Meaning and purpose
β (total, 6 groups)	dimensionless	Total delayed-neutron fraction from evaluated U-235 thermal-fission data. Sets the scale of the dollar unit and the margin to prompt criticality; without delayed neutrons a reactor would be uncontrollable.
λ_i precursors	6 values, s^{-1}	Decay constants of the six delayed-neutron precursor groups (U-235). Together with the group fractions β_i they determine the time response of the reactor — the inhour-equation period for any reactivity insertion follows from these.
$\lambda(I-135)$	s^{-1}	Iodine-135 decay constant (half-life ≈ 6.6 h). I-135 is the precursor of Xe-135; this constant sets the delay between a power change and the resulting xenon response.
$\lambda(Xe-135)$	s^{-1}	Xenon-135 decay constant (half-life ≈ 9.1 h). Together with $\lambda(I-135)$ it shapes the post-shutdown xenon peak and the return to equilibrium.
$\gamma(I-135)$	dimensionless	Cumulative fission yield of I-135 — the fraction of fissions producing an I-135 atom. The dominant source term of the xenon chain.
$\gamma(Xe-135)$	dimensionless	Direct fission yield of Xe-135 (the small fraction born directly in fission, in addition to the iodine decay path).

2.2 · Model parameters (representative choices)

Parameter	Value / form	Meaning and purpose
Λ (generation time)	μs	Mean neutron generation time — the prompt timescale of the kinetics. A representative PWR-order value; it dominates the very fast response (and the period at large reactivity), but barely affects delayed-neutron-dominated transients.
α_D (Doppler)	pcm/ $^{\circ}C$	Fuel-temperature (Doppler) reactivity coefficient. Negative: as fuel heats, U-238 resonance absorption broadens and reactivity falls. The fastest-acting inherent safety feedback in the model — it is what makes the simulated reactor self-regulate after a power increase.
α_M (moderator)	pcm/ $^{\circ}C$	Moderator (coolant) temperature coefficient. Negative: hotter, less dense water moderates less effectively. Held constant here — flagged honestly on screen, because a real MTC is state-dependent (boron, burnup, temperature).
Xe-135 equiv. worth	pcm	The lumped reactivity worth assigned to equilibrium xenon. Converts the dimensionless Xe concentration into pcm of negative reactivity, scaling the whole poison feedback including the depth of the post-shutdown pit.

Parameter	Value / form	Meaning and purpose
Xe burnout $\sigma\phi$	s^{-1}	Lumped product of the Xe-135 absorption cross-section and the neutron flux at full power. Governs how fast xenon is burned out by the flux — the term that makes xenon behaviour power-dependent. Explicitly not a microscopic cross-section.
Thermal gains K_P, K_FC, K_CO	<i>lumped</i>	Gains of the two-node (fuel/coolant) thermal model: K_P converts power to heat input, K_FC couples fuel to coolant, K_CO couples coolant to the heat sink. Hand-tuned so the model reproduces the declared reference operating point — they are the bridge between neutronics and temperature feedback.
Heat capacities C_F, C_C	<i>lumped</i>	Lumped heat capacities of the fuel and coolant nodes. Set the thermal time constants — how quickly temperatures (and therefore the feedback reactivities) respond to a power change.
Reference temps T_F0 / T_C0 / T_in	$^{\circ}\text{C}$	Fuel, coolant and inlet temperatures at the 100%-power operating point. The anchor of the thermal model: feedback reactivity is computed from deviations relative to these references, and the thermal-equilibrium validation check verifies the gains reproduce them.

These tables are read directly from the running solver, so the simulation, the on-screen table and the JSON export are provably the same numbers. The JSON export additionally includes the per-group β_i values, the fission-source scaling and the 12-group decay-heat fractions and time constants used by the afterheat model.

3 · Operating margins — live, demonstrator setpoints

Live distances between the current plant state and illustrative trip setpoints. Each row shows the value, the limit, the remaining margin and a status (HEALTHY / WATCH / MARGINAL).

Parameter	Value / form	Meaning and purpose
Reactor power	$\% \text{ rated} \cdot \text{limit} \leq 118\%$	Current power against an overpower-trip setpoint. Purpose: the most basic protection margin — how far the unit is from a high-power trip.
Reactor period	$s \cdot \text{limit} \geq 10 \text{ s}$	The e-folding time of power when rising. A short positive period means power is climbing fast; the margin is the distance to a short-period trip. Shown as 'stable' when effectively infinite and N/A when power is decreasing (a negative period is not a protection concern).
Avg fuel temp	$^{\circ}\text{C} \cdot \text{limit} \leq 700^{\circ}\text{C}$	Average fuel-node temperature against a representative band. Purpose: a stand-in for fuel thermal limits in a model that has only one lumped fuel node.
Reactivity	$\$ \cdot \text{limit} < \1.00	Net reactivity expressed in dollars (ρ/β). Purpose: the margin to prompt criticality — the hardest boundary in reactor kinetics. Crossing \$1.00 would make the chain reaction self-sustaining on prompt neutrons alone.

Why DNBR and linear heat rate are deliberately absent

They are thermal-hydraulic outputs that require a subchannel-flow model, a critical-heat-flux correlation and an axial power distribution. NEXUS-1 is a point-kinetics model with a single lumped thermal node, so it cannot compute them — displaying a fabricated DNBR would misrepresent the model. The setpoints shown are demonstrator values, not real Technical Specification limits.

4 · Sensitivity & uncertainty propagation

Answers the question every reviewer asks of a model: which inputs actually matter? Each parameter is perturbed $\pm 1\%$ and the model re-run (central difference), giving the normalized, dimensionless sensitivity $S = (p/Y) \cdot (\partial Y / \partial p)$ — legitimately rankable across parameters with different units.

Parameter	Value / form	Meaning and purpose
Asymptotic period (+10 pcm)	<i>output</i> · s	How fast the reactor responds to a small reactivity insertion. Sensitive almost only to Λ and β — the pure-kinetics constants.

Parameter	Value / form	Meaning and purpose
Equilibrium fuel temp	output · °C	The steady-state fuel temperature at 100% power. Driven by the thermal gains (K_P, K_FC, K_CO) and the inlet temperature — the thermal half of the model.
Xenon pit depth (post-SCRAM)	output · pcm	The reactivity deficit at the post-shutdown xenon peak. Driven by the xenon-chain constants (λ_I , λ_{Xe} , σ_ϕ , γ_I , γ_{Xe}) and the lumped Xe worth.
Input uncertainties σ_p	$\pm 2\%$ / $\pm 10\text{--}15\%$	The assumed relative uncertainty of each input: nuclear data $\pm 2\%$ (well-evaluated quantities), modeling parameters $\pm 10\text{--}15\%$ (lumped, representative choices). Explicitly labelled illustrative.
Propagated 2σ bands	per output	Output uncertainty from first-order propagation, $\sigma_{Y^2} \approx \sum (\partial Y / \partial p_i)^2 \cdot \sigma_{p,i}^2$ — the model-form half of an uncertainty estimate, distinct from sensor/measurement noise.

The deliberate take-away shown on screen: the dominant parameter differs by output — Λ and β drive the period, the thermal gains drive temperature, the xenon constants drive the pit. There is no single 'most influential parameter'.

5 · Xenon forecast — forward integration from live state

Parameter	Value / form	Meaning and purpose
Current Xe-135	% of equilibrium	The live xenon concentration relative to its full-power equilibrium value — the model's poison state right now, normalised so it is comparable across units.
Trend at current power	rising / falling / flat	The sign of dXe/dt evaluated from the live state: whether xenon is building, decaying, or near equilibrium for the current power.
Projected peak	% of eq · hours	The maximum of a 48-hour forward integration of the coupled I-135/Xe-135 equations with power held constant — when the peak arrives and how high it is. After a SCRAM this is the classic xenon pit forecast.
Illustrative 110% band	hours to crossing	If the projection crosses 110% of equilibrium, the time to that crossing is shown. An illustrative awareness band, explicitly not a Technical Specification limit.

This is a true forward integration, not a linear extrapolation — xenon rises, turns over and peaks, so a slope projection would miss the peak entirely. A change in power would alter the trajectory; no control action is prescribed.

Verifiability

The Parameters (JSON) export on the Model Analysis screen serialises every constant in section 2 directly from the running solver, and the Live trace (CSV) export captures the live state — so every claim in this document can be checked against the running application with independent tools.